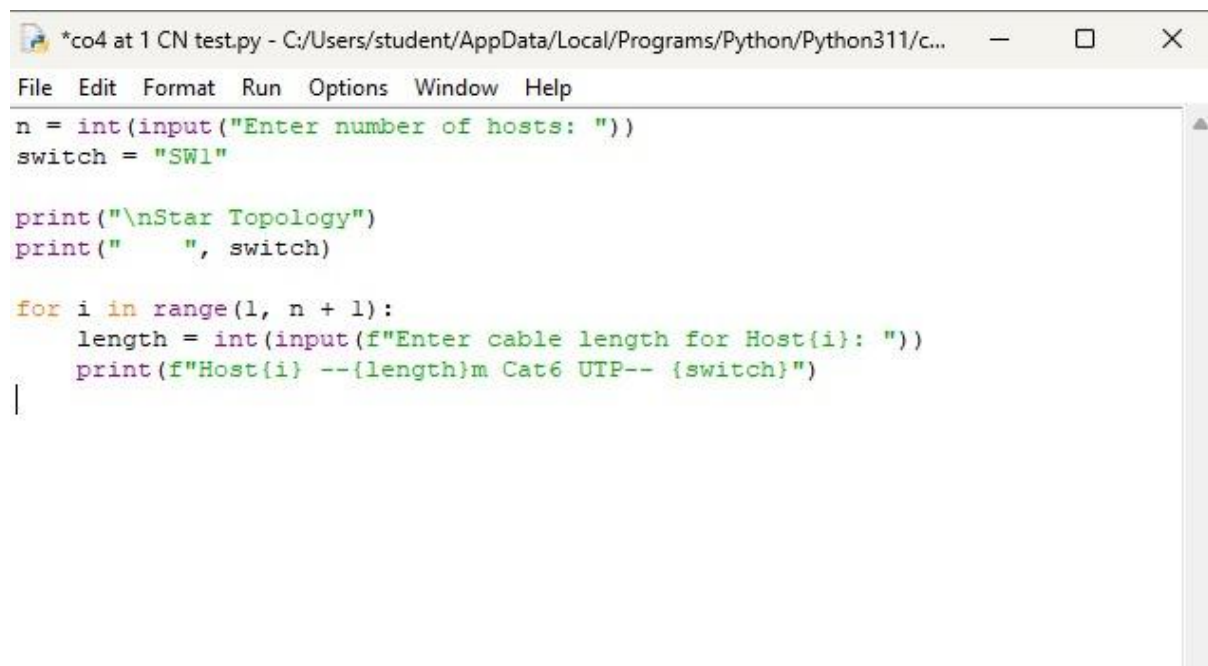


1.

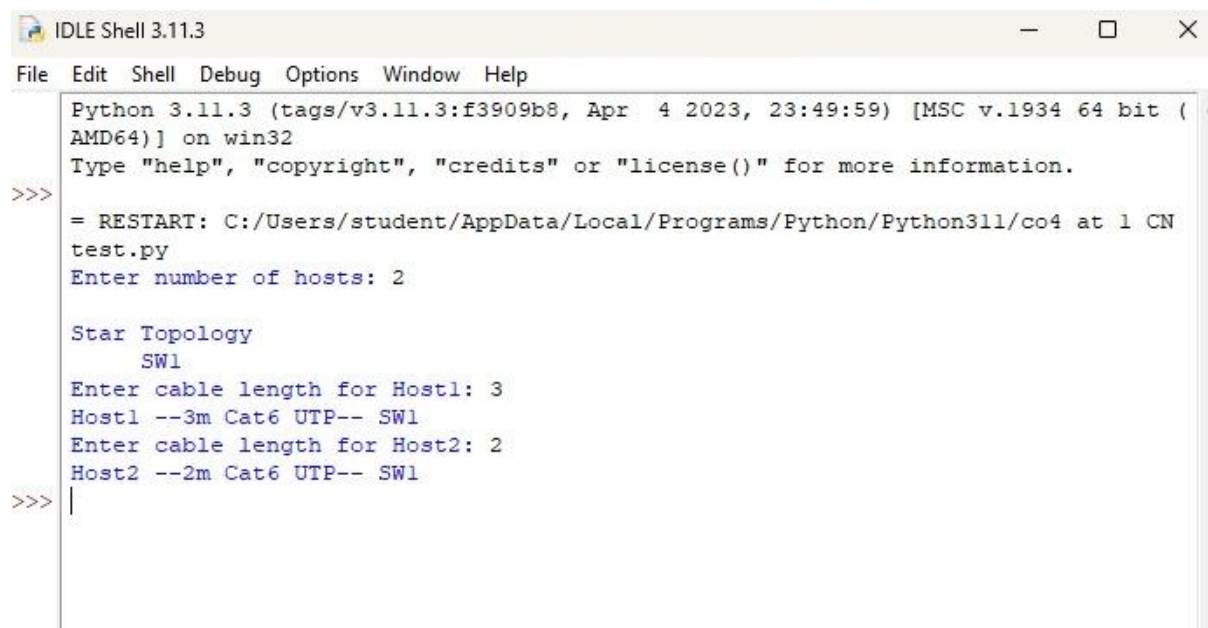


```
*co4 at 1 CN test.py - C:/Users/student/AppData/Local/Programs/Python/Python311/c...
File Edit Format Run Options Window Help
n = int(input("Enter number of hosts: "))
switch = "SW1"

print("\nStar Topology")
print("    ", switch)

for i in range(1, n + 1):
    length = int(input(f"Enter cable length for Host{i}: "))
    print(f"Host{i} --{length}m Cat6 UTP-- {switch}")
|
```

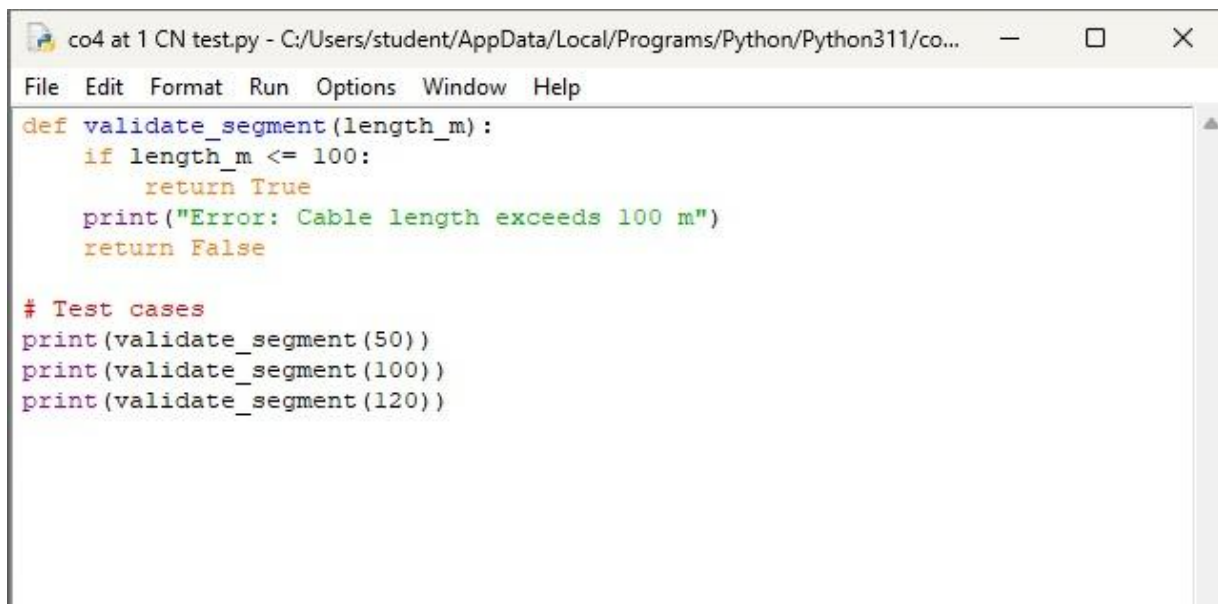
Output:



```
IDLE Shell 3.11.3
File Edit Shell Debug Options Window Help
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/student/AppData/Local/Programs/Python/Python311/co4 at 1 CN test.py
Enter number of hosts: 2

Star Topology
    SW1
Enter cable length for Host1: 3
Host1 --3m Cat6 UTP-- SW1
Enter cable length for Host2: 2
Host2 --2m Cat6 UTP-- SW1
>>> |
```

2.



```
co4 at 1 CN test.py - C:/Users/student/AppData/Local/Programs/Python/Python311/co...
File Edit Format Run Options Window Help

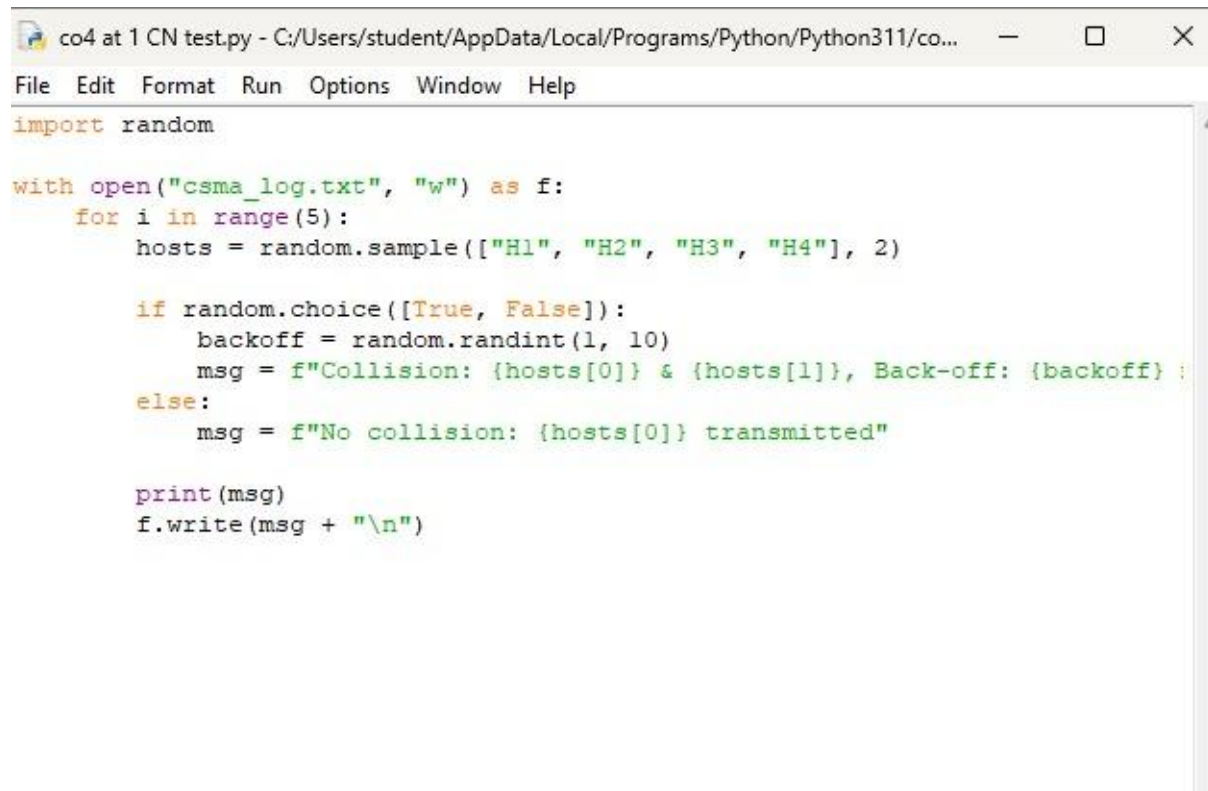
def validate_segment(length_m):
    if length_m <= 100:
        return True
    print("Error: Cable length exceeds 100 m")
    return False

# Test cases
print(validate_segment(50))
print(validate_segment(100))
print(validate_segment(120))
```

Output:

```
= RESTART: C:/Users/student/AppData/Local/Programs/Python/Python311/co4
at 1 CN test.py
True
True
Error: Cable length exceeds 100 m
False
|
```

3.



```
co4 at 1 CN test.py - C:/Users/student/AppData/Local/Programs/Python/Python311/co...
File Edit Format Run Options Window Help

import random

with open("csma_log.txt", "w") as f:
    for i in range(5):
        hosts = random.sample(["H1", "H2", "H3", "H4"], 2)

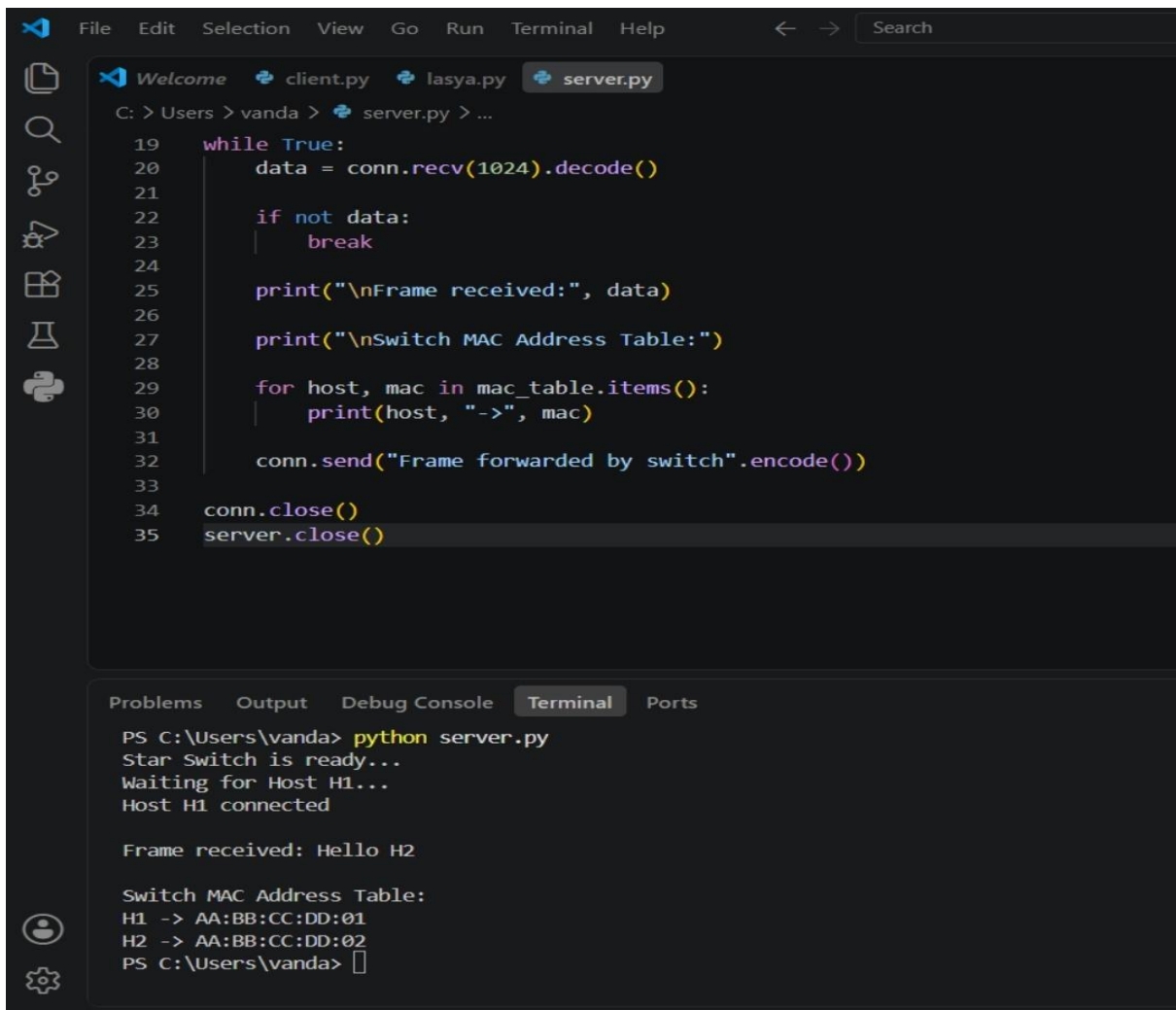
        if random.choice([True, False]):
            backoff = random.randint(1, 10)
            msg = f"Collision: {hosts[0]} & {hosts[1]}, Back-off: {backoff} !"
        else:
            msg = f"No collision: {hosts[0]} transmitted"

        print(msg)
        f.write(msg + "\n")
```

Output:

```
= RESTART: C:/Users/student/AppData/Local/Programs/Python/Python311/co4
at 1 CN test.py
No collision: H3 transmitted
No collision: H2 transmitted
No collision: H4 transmitted
No collision: H2 transmitted
Collision: H2 & H1, Back-off: 4 ms
```

4.



The screenshot shows the Visual Studio Code editor with the `server.py` file open. The code is a Python script that listens for incoming connections on `localhost`. It receives a message from `H2` and prints the switch MAC address table. The terminal output shows the execution of `python server.py`, which prints "Star Switch is ready...", "Waiting for Host H1...", "Host H1 connected", "Frame received: Hello H2", and the "Switch MAC Address Table" with entries for H1 and H2.

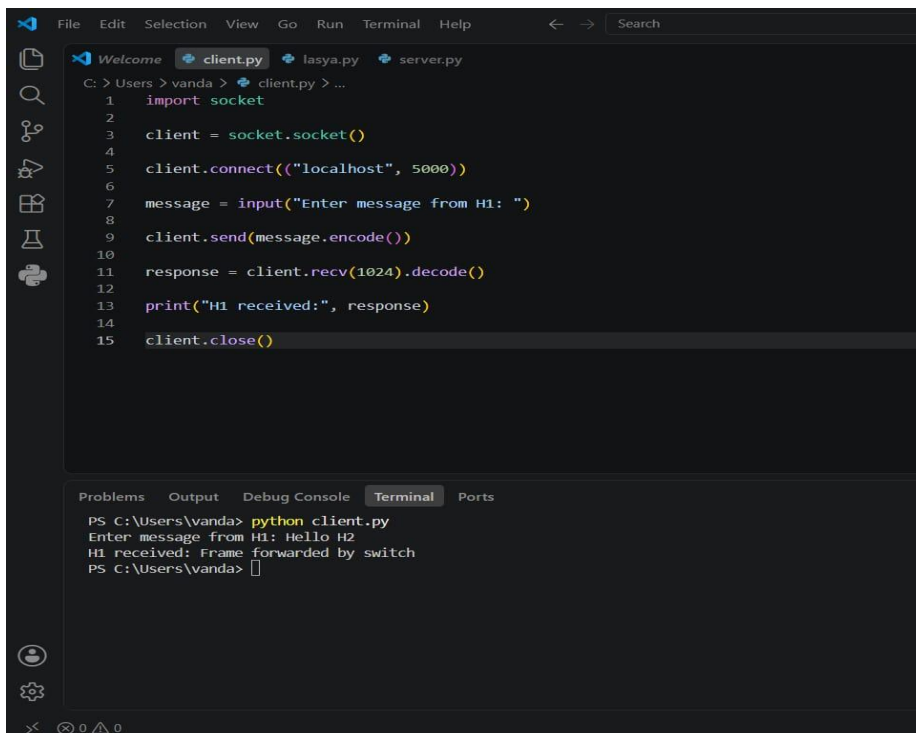
```
File Edit Selection View Go Run Terminal Help
Welcome client.py lasya.py server.py
C: > Users > vanda > server.py > ...
19 while True:
20     data = conn.recv(1024).decode()
21
22     if not data:
23         break
24
25     print("\nFrame received:", data)
26
27     print("\nSwitch MAC Address Table:")
28
29     for host, mac in mac_table.items():
30         print(host, "->", mac)
31
32     conn.send("Frame forwarded by switch".encode())
33
34 conn.close()
35 server.close()
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\vanda> python server.py
Star Switch is ready...
Waiting for Host H1...
Host H1 connected

Frame received: Hello H2

Switch MAC Address Table:
H1 -> AA:BB:CC:DD:01
H2 -> AA:BB:CC:DD:02
PS C:\Users\vanda>
```



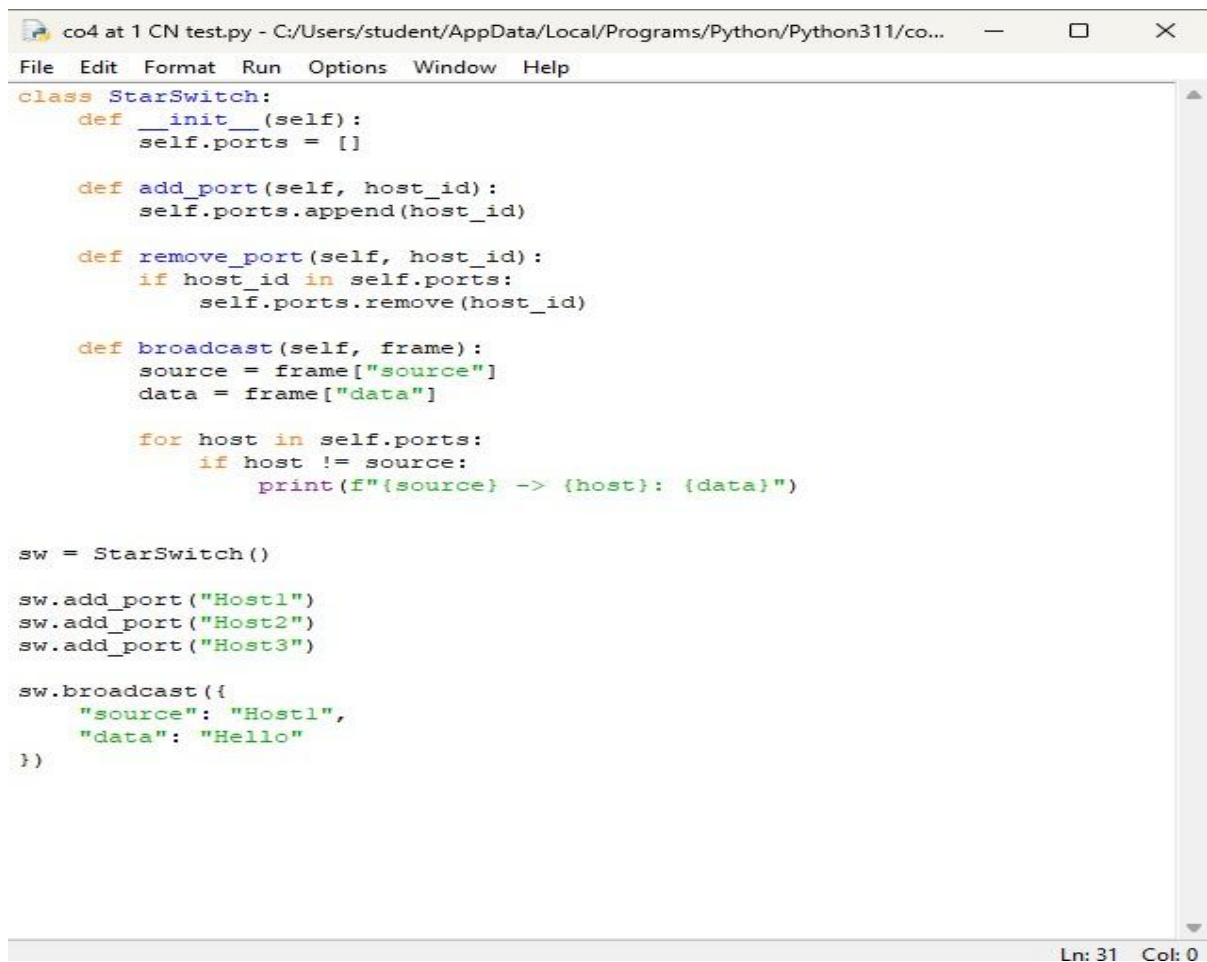
The screenshot shows the Visual Studio Code editor with the `client.py` file open. The code is a Python script that connects to `localhost` on port `5000`. It sends a message from `H1` and receives a response from `H2`. The terminal output shows the execution of `python client.py`, which prompts for a message from `H1` and prints the received response.

```
File Edit Selection View Go Run Terminal Help
Welcome client.py lasya.py server.py
C: > Users > vanda > client.py > ...
1 import socket
2
3 client = socket.socket()
4
5 client.connect(("localhost", 5000))
6
7 message = input("Enter message from H1: ")
8
9 client.send(message.encode())
10
11 response = client.recv(1024).decode()
12
13 print("H1 received:", response)
14
15 client.close()
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\vanda> python client.py
Enter message from H1: Hello H2
H1 received: Frame forwarded by switch
PS C:\Users\vanda>
```

5.



```
co4 at 1 CN test.py - C:/Users/student/AppData/Local/Programs/Python/Python311/co...
File Edit Format Run Options Window Help

class StarSwitch:
    def __init__(self):
        self.ports = []

    def add_port(self, host_id):
        self.ports.append(host_id)

    def remove_port(self, host_id):
        if host_id in self.ports:
            self.ports.remove(host_id)

    def broadcast(self, frame):
        source = frame["source"]
        data = frame["data"]

        for host in self.ports:
            if host != source:
                print(f"{source} -> {host}: {data}")

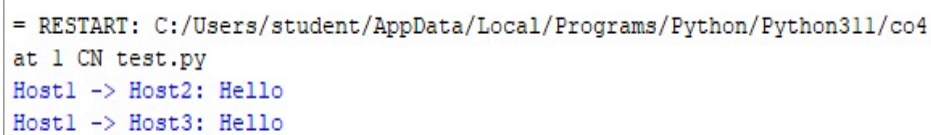
sw = StarSwitch()

sw.add_port("Host1")
sw.add_port("Host2")
sw.add_port("Host3")

sw.broadcast({
    "source": "Host1",
    "data": "Hello"
})

Ln: 31 Col: 0
```

Output:



```
= RESTART: C:/Users/student/AppData/Local/Programs/Python/Python311/co4
at 1 CN test.py
Host1 -> Host2: Hello
Host1 -> Host3: Hello

Ln: 35 Col: 0
```